

1.) The smallest division on the main scale of a Vernier Caliper

- A. mm
- B. cm
- C. m
- D. micrometer

Answer: A

2.) The smallest division on the sleeve of a Micrometer Caliper

- A. mm
- B. cm
- C. m
- D. micrometer

Answer: A

3.) Which part of the Vernier caliper is primarily used to measure the internal diameter of a pipe or cylinder?

- A. main scale
- B. depth measuring blade
- C. upper jaws
- D. lower jaws

Answer: C

4.) Which dimensions can a standard vernier caliper measure?

- A. Outer only
- B. Outer and Inner only
- C. Weight and volume
- D. Outer, Inner, and depth only

Answer: D

5.) The main scale reading is 2.5 cm, the coinciding vernier division is 7. What is the reading?

- A. 2.57 mm
- B. 2.57 cm
- C. 25.7 cm
- D. 2.57 m

Answer: B

6.) The “anvil” of a micrometer is the movable part that clamps onto the measured object.

- A. TRUE
- B. FALSE

Answer: B

7.) Suppose you have two forces applied to a central ring on the force table: $F_1 = 4.9\text{N}$ @ 30° , $F_2 = 8\text{N}$ @ 120° . Find the magnitude of the equilibrant.

- A. 9.95N
- B. 8.87N
- C. 10.96N
- D. 11.55N

Answer: C

8.) Refer to question 7. What is the direction of the equilibrant?

- A. 93.5°
- B. 273.5°
- C. 125.5°
- D. 205°

Answer: D

9.) Using the force table, the three vectors were balanced by placing counterweights on the fourth unused string and locating the appropriate angle or direction. These counterweights on the fourth position represent what vector?

- A. resultant
- B. equilibrant
- C. total vector
- D. total scalar

Answer: B

10.) The tip-tail method is also called _____ method.

- A. force table
- B. polygon
- C. parallelogram
- D. triangulation

Answer: B

11.) What is the unit of measurement for force?

- A. Newton
- B. Kilograms
- C. Joules
- D. meter

Answer: A

12.) What is the purpose of the Force Table Physics Lab 1?

- A. To study the effects of forces on an object
- B. To investigate the concept of equilibrium
- C. To measure the acceleration of an object
- D. To analyze the motion of a projectile

Answer: B

13.) In Activity 3: Motion Down the Incline, as the height of elevation of the track increases, the acceleration of the cart also increases while its mass remain constant.

- A. TRUE
- B. FALSE
- C. cannot be determined

Answer: A

14.) In Activity 3: Motion Down the Incline, as the mass increases, its acceleration also increases while the height of elevation of the track remains constant.

- A. TRUE
- B. FALSE
- C. cannot be determined

Answer: B

15.) What quantity does the slope of the best-fit line described by the acceleration a versus h/L graph represent?

- A. instantaneous velocity
- B. gravity
- C. average speed
- D. free-fall

Answer: B

16.) In PASCO Capstone, the position-time graph of a cart moving down the incline forms a curve opening upward. What does this indicate?

- A. constant velocity
- B. instantaneous velocity
- C. average acceleration
- D. increasing velocity

Answer: D

17.) A cart’s velocity-time graph is a straight line with positive slope. The slope represents:

- A. instantaneous velocity
- B. acceleration
- C. average speed
- D. free-fall

Answer: B

18.) The theoretical acceleration is 1.50 m/s² while the measured acceleration is 1.35 m/s². Find the percent error.

- A. 15%
- B. 10%
- C. 12%
- D. 5%

Answer: B

19.) Two projectiles are launched at the same speed. Projectile A: 30° Projectile B: 60° Which stays longer in the air?

- A. Projectile A
- B. Projectile B

Answer: B

20.) Refer to question 19, which traveled farther

- A. Projectile A
- B. Projectile B

Answer: A and B

Same speed, angles 30° and 60°
Range formula: $R = v^2 \sin(2\theta) / g$

Projectile A: 30°
 $\sin(2 \times 30) = \sin 60^\circ$

Projectile B: 60°
 $\sin(2 \times 60) = \sin 120^\circ$

 $\sin 60^\circ = \sin 120^\circ$, so same distance sila.

21.) What is the horizontal acceleration on projectiles.

- A. Depends on the force on the ball
- B. zero
- C. 9.81 m/s²
- D. Cannot be determined

Answer: B

22.) What is the vertical acceleration on projectiles

- A. Depends on the force on the ball
- B. zero
- C. 9.81 m/s²
- D. Cannot be determined

Answer: C

23.) What is the ball’s vertical velocity on the peak of its path

- A. Depends on the force on the ball
- B. zero
- C. 9.81 m/s²
- D. Cannot be determined

Answer: B

24.) In a projectile lab, a ball lands 2.4 m away after 0.4 s. Find the horizontal velocity.

- A. 4m/s
- B. 5m/s
- C. 6m/s
- D. 7m/s

Answer: C

25.) A grocery cart accelerates 5 m/s² when a push, a force of 10 Newtons is applied to it. What is the mass of the object, in kg?

- A. 2
- B. 5
- C. 3
- D. 4

Answer: A

26.) Newton’s First Law of Motion

- A. Law of Inertia
- B. Law of Acceleration
- C. Law of Reaction
- D. Law of Conservation of Mass

Answer: A

27.) A force of 20 N acts on a 4 kg object. Find the acceleration (m/s²)

- A. 2
- B. 3
- C. 4
- D. 5

Answer: D

28.) A truck has a mass of 2000 kg. An engine provides a force of 4000 N. Find the acceleration. (m/s²)

- A. 2
- B. 3
- C. 4
- D. 5

Answer: A

29.) What happens to the acceleration of an object if the net force acting on it doubles while the mass stays the same?

- A. doubles
- B. quadruples
- C. halved
- D. x1.5

Answer: A

30.) What happens to acceleration if the mass of an object doubles while the force remains constant?

- A. doubles
- B. quadruples
- C. halved
- D. x1.5

Answer: C